

CH 12.53

DATA SET

DS1210

9. Lots of Highways The number of miles of U.S. urban roadways (millions of miles) for the years 2000–2015 is reported below.⁶ The years are simplified as years 0 through 15.

Miles of Roadways (millions)	0.85	0.88	0.89	0.94	0.98	1.01	1.03	1.04
Year – 2000	0	1	2	3	4	5	6	7

Miles of Roadways (millions)	1.07	1.08	1.09	1.10	1.11	1.18	1.20	1.21
Year – 2000	8	9	10	11	12	13	14	15

- Draw a scatterplot of the number of miles of roadways in the U.S. over time. Describe the pattern that you see.
- Find the least-squares line describing these data. Do the data indicate that there is a linear relationship between the number of miles of roadways and the year? Test using a t statistic with $\alpha = .05$.
- Construct the ANOVA table and use the F statistic to answer the question in part b. Verify that the square of the t statistic in part b is equal to F .
- Calculate r^2 . What does this value tell you about the effectiveness of the linear regression analysis?

$$c. \quad SS = S_{yy} = \sum y^2 - n \bar{y}^2 = 0.186$$

$$SSR = \frac{S_{xy}^2}{S_{xx}} = \frac{(\sum xy - n \bar{x} \bar{y})^2}{\sum x^2 - n \bar{x}^2} = 0.181$$

$$SSE = SS - SSR = 0.00476$$

ANOVA	SS	df	MS	F
SSR	0.181	1	0.181	502.78
SSE	0.005	14	0.00036	

$$SS = 0.186 \quad 15$$

H_0 : model is not useful in explaining y .

H_a : model is useful in explaining y .

$$F_{STAT} = \frac{MSR}{MSE} \sim F_{1,14}$$

$$F^* = 502.78$$

$$RR: \{F \mid F > F_{1,14}\} = \{F \mid F > 4.6\}$$

$$502.78 > 4.6 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ We have sufficient evidence to say that the model is useful in explaining y .

$$b. \quad \hat{y} = a + bx$$

$$b = \frac{S_{xy}}{S_{xx}} = \frac{\sum xy - n \bar{x} \bar{y}}{\sum x^2 - n \bar{x}^2} = \frac{7.85}{340} = 0.0231$$

$$a = \bar{y} - b \bar{x} = 0.868 \quad \hat{y} = 0.868 + 0.0231x$$

$$H_0: \beta = 0, H_a: \beta \neq 0$$

$$\alpha = 0.05$$

$$t_{STAT} = \frac{b - 0}{\sqrt{\frac{MSE}{S_{xx}}}} \sim t_{15}$$

$$t^* = \frac{0.0231}{\sqrt{\frac{0.00036}{340}}} = 22.45$$

$$MSE = \frac{SS - SSR}{n - 2} = \frac{3yy - \frac{S_{xy}^2}{S_{xx}}}{n - 2} = 0.00036$$

$$t_{14, 0.025} = 2.145 \quad RR: \{t \mid |t| > 2.145\}$$

$$22.45 > 2.145 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ there's sufficient evidence to say that there is a linear relationship.

$$d. \quad r^2 = 0.972$$

$\Rightarrow$ model 可解釋的資料變異比例 = 97.2%

$\Rightarrow$ model effective